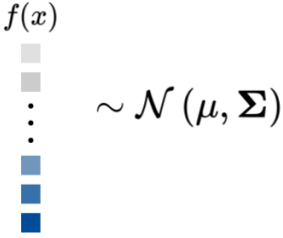
Introduction to Machine Learning

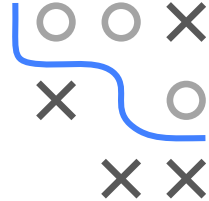
Gaussian Processes
Stochastic Processes and Distributions on Functions



Learning goals

- GPs model distributions over functions
- The marginalization property makes this distribution easily tractable
- GPs are fully specified by mean and covariance function
- GPs are indexed families

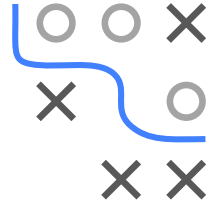
WEIGHT-SPACE VIEW



- Until now: hypothesis space $\mathcal{H}$ of parameterized functions $f(\mathbf{x} \mid \boldsymbol{\theta})$
- ERM: find risk-minimal parameters (weights) $\boldsymbol{\theta}$
- Bayesian paradigm: distribution over $\boldsymbol{\theta} \Rightarrow$ update prior to posterior belief after observing data acc to Bayes' rule

$$p(\boldsymbol{\theta}|\mathbf{X}, \mathbf{y}) = \frac{\text{likelihood} \cdot \text{prior}}{\text{marginal likelihood}} = \frac{p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) \cdot q(\boldsymbol{\theta})}{p(\mathbf{y}|\mathbf{X})}$$

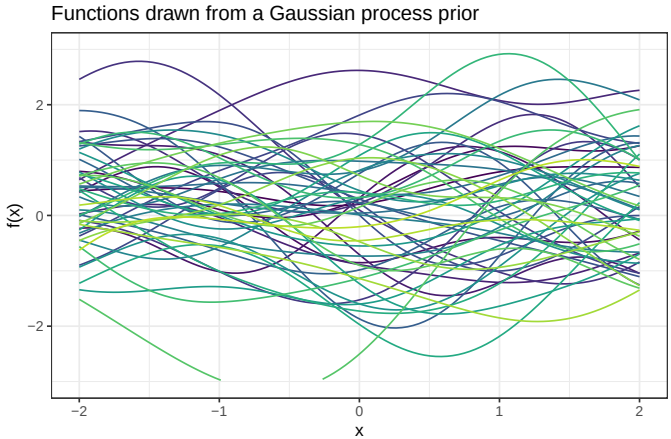
FUNCTION-SPACE VIEW



- New POV: rather than finding $\boldsymbol{\theta}$ to parameterize $f(\mathbf{x} \mid \boldsymbol{\theta})$ in parameter space, search space of admissible functions directly
- Sticking to Bayesian inference, specify prior distribution **over functions** and update according to observed data points

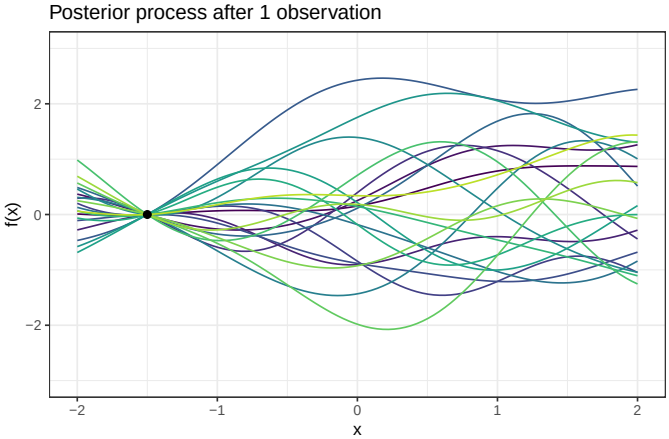
DRAWING FROM FUNCTION PRIORS

- Imagine we could draw functions from some prior distribution



DRAWING FROM FUNCTION PRIORS

- After observing some data points, we restrict sampling to functions consistent with the data



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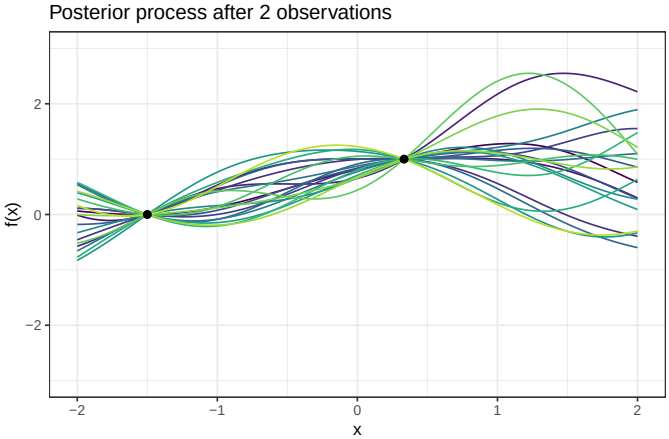

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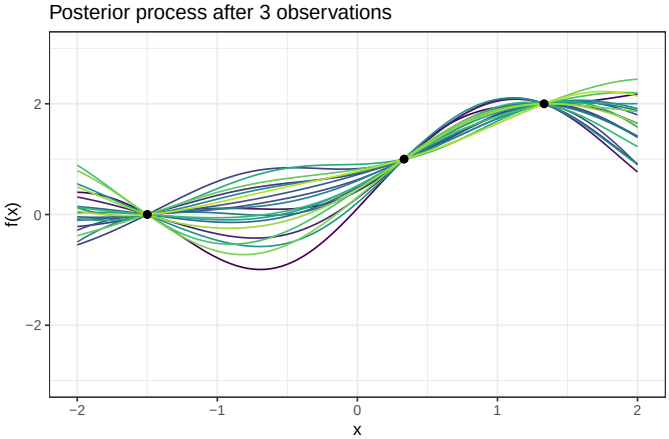
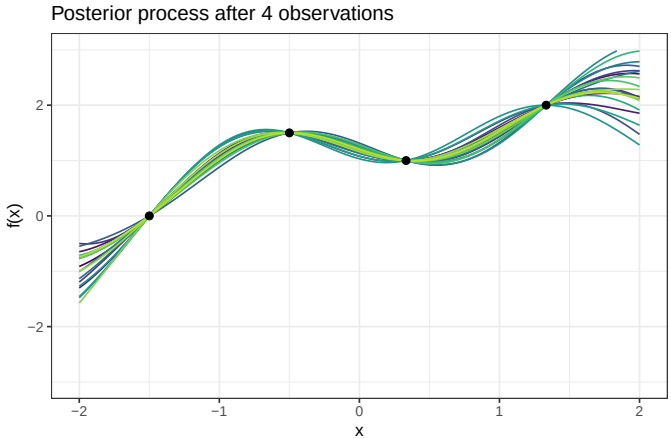


DRAWING FROM FUNCTION PRIORS

- Variety of admissible functions shrinks with observing more data



- Intuitively: distributions over functions have “mean” & “variance”

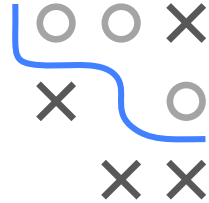
WEIGHT-SPACE VS. FUNCTION-SPACE VIEW

Weight-Space View

- Parameterize functions (e.g., $f(\mathbf{x} \mid \boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{x}$)
- Define distributions on $\boldsymbol{\theta}$
- Inference in param space Θ

Function-Space View

- Work on functions directly
- Define distributions on f
- Inference in fun space $\mathcal{H}$

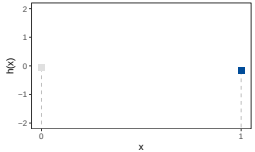


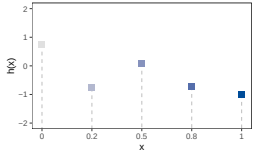
DISCRETE FUNCTIONS

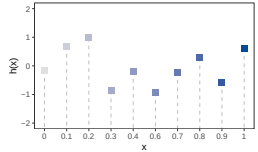
- Let $\mathcal{X} = \left\{\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)}\right\}$, $\mathcal{H} = \{h \mid h : \mathcal{X} \to \mathbb{R}\}$
- Any $h \in \mathcal{H}$ has finite domain with $n < \infty$ elements
  $\Rightarrow$ neat representation with n-dim vector

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \dots, h(\mathbf{x}^{(n)})]$$

- Example functions living in this space for $|\mathcal{H}| \in \{2, 5, 10\}$

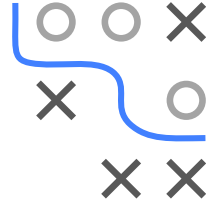






TODO LW: new plots

DISTRIBUTIONS ON DISCRETE FUNCTIONS

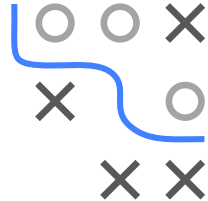


- Specify (discrete) probability function on functions $h \in \mathcal{H}$
- Natural way: vector representation as n-dim RV, e.g.,

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- For now: set $\mathbf{m} = \mathbf{0}$, assume $\mathbf{K}$ to be given

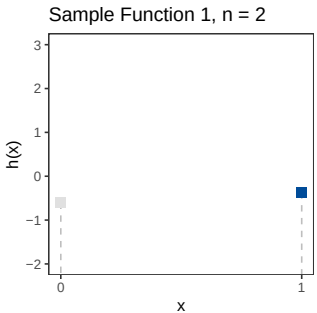
EXAMPLE: RANDOM DISCRETE FUNCTIONS

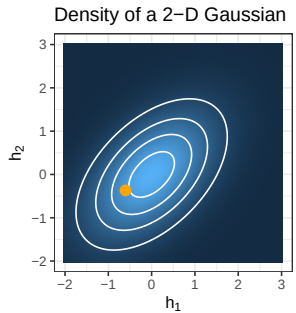


- Example ctd: $h : \mathcal{X} \to \mathcal{Y}$ defined on 2 points
- Sample representatives by sampling from a 2-dim Gaussian

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), h(\mathbf{x}^{(2)})] \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$$

- Samples: weakly ($\mathbf{K} \approx \boldsymbol{I}$) correlated components





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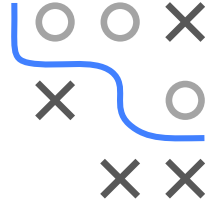

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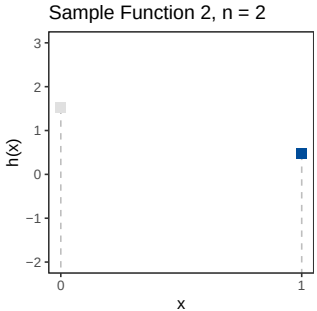
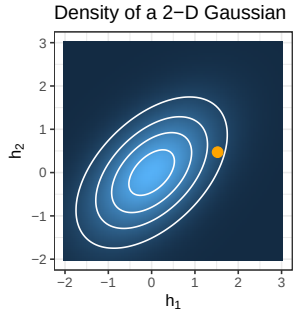
- Samples: strongly (large off-diag in $\mathbf{K}$) correlated components





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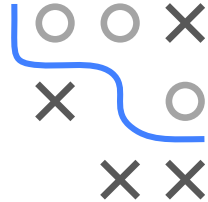

- Example ctd: $h : \mathcal{X} \to \mathcal{Y}$ defined on 5 points
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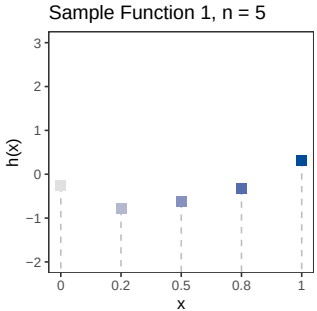

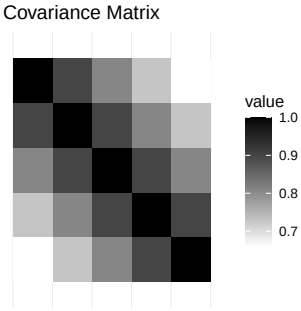

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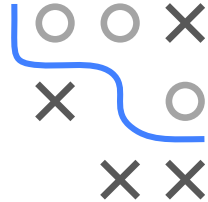

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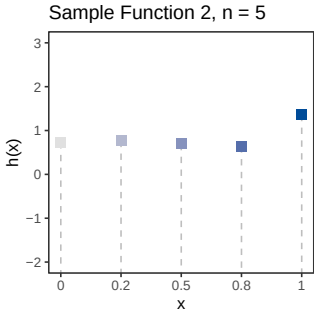

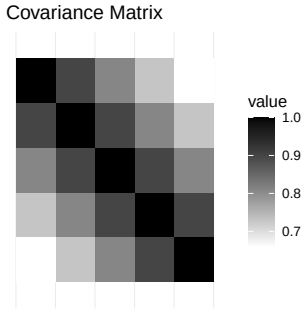

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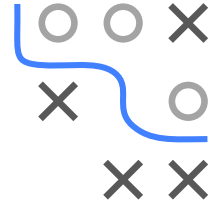
SPATIAL CORRELATION



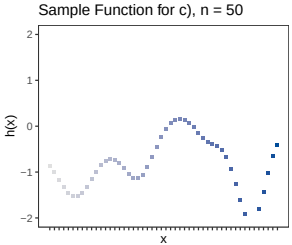
- "Meaningful" functions (on numeric $\mathcal{X}$) often have spatial property:

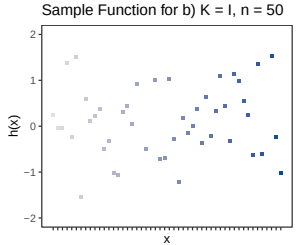
$$\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \text{ close in } \mathcal{X} \Rightarrow f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)}) \text{ close in } \mathcal{Y}$$

- In other words: fun values of nearby points should be correlated
- Enforce this by choosing dist-based covariance function

$$\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \text{ close in } \mathcal{X} \Leftrightarrow \mathbf{K}_{ij} \text{ high}$$

- E.g., $\mathbf{K}_{ij} = k(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) = \exp\left(-\frac{1}{2}\|\mathbf{x}^{(i)} - \mathbf{x}^{(j)}\|^2\right)$





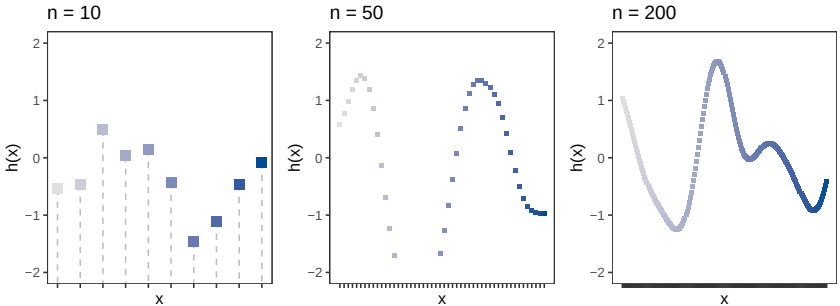
- More on covariance function, or **kernel**, $k(\cdot, \cdot)$ later on TODO LW: new plots (adapt titles)

FROM DISCRETE TO CONTINUOUS FUNCTIONS

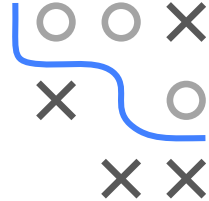
- Recall: we defined distributions on functions by (Gaussian) random vector of corresponding function values

$$\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- We can do this for ever larger n (as "granular" as we want)



FROM DISCRETE TO CONTINUOUS FUNCTIONS



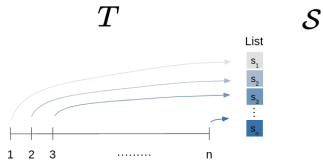
- No matter how large n is: still functions over discrete domains
- Can we simply extend our distribution def to **continuous**-domain functions by taking $n \to \infty$?
- Unclear how to obtain "infinitely" long (Gaussian) random vectors
- Observation: random vectors $\boldsymbol{h} = [h(\mathbf{x}^{(1)}), \ldots, h(\mathbf{x}^{(n)})]$ are collections of RVs enumerated by $\{1, \ldots, n\} \Rightarrow$ **indexed family**
- Can we use more general, infinite index sets?

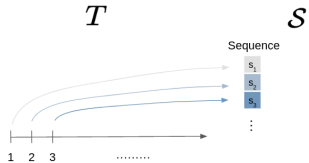
DEFINITION: INDEXED FAMILY

- Mathematical “rule” to map indices to some state space

$$s : T \to \mathcal{S}, \quad t \mapsto s_t = s(t)$$

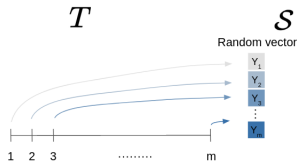
- Gives rise to collection $\{s_t : t \in T\}$
- Can assume many forms depending on T and $\mathcal{S}$
- Example: real-valued $\mathcal{S}$
  - $\mathcal{S} = \mathbb{R}$, $t \mapsto s_t$
  - Finite index set, e.g., $T = \{1, \ldots, n\} \Rightarrow$ list
  - Countably ∞ index set, e.g., $T = \mathbb{N} \Rightarrow$ sequence
  - Uncountably ∞ index set, e.g., $T = \mathbb{R} \Rightarrow$ example: $\{\sin(t) : t \in \mathbb{R}\}$

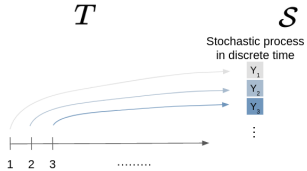




DEFINITION: STOCHASTIC PROCESS

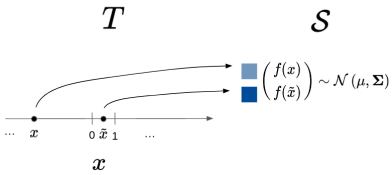
- Special kind of indexed family: RVs with specific distributional assumption + cov structure $\Rightarrow \{Y_t : t \in T\}$
- Intuition: probability distributions describe random vectors, SP describe random functions
- Examples
  - $\mathcal{S}$: space of RVs, $t \mapsto Y_t$
  - Finite index set, e.g., $T = \{1, \ldots, n\}$ $\Rightarrow$ random vector
  - Countably ∞ index set, e.g., $T = \mathbb{N} \Rightarrow$ discrete-time SP
  - Uncountably ∞ index set, e.g., $T = \mathbb{R} \Rightarrow$ continuous-time SP

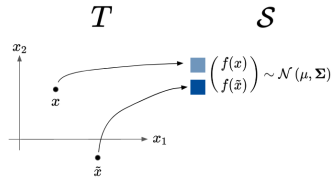




INTUITION: GAUSSIAN PROCESSES

- Special kind of SP
- Index set: $\mathcal{X}$, distr assumption + cov structure: Gaussian
- Corollary: any finite random vector drawn from a GP is multivariate Gaussian $\Rightarrow$ **marginalization property**

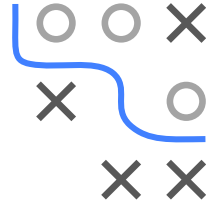




- Precisely what we were looking for: process to generate arbitrarily long Gaussian random vectors!

MARGINALIZATION PROPERTY

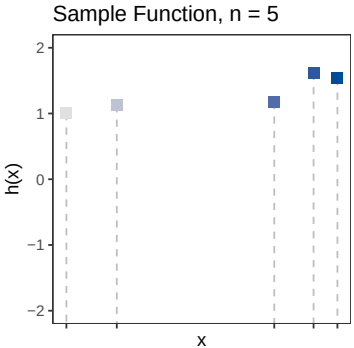
- Consider finite set of inputs $\boldsymbol{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\} \subset \mathcal{X}$



- Distinguishing feature of GPs: for any such $\boldsymbol{X}$,

$$\mathbf{f} = f(\boldsymbol{X}) = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- $\mathbf{m}$ and $\mathbf{K}$ are calculated by a mean and cov function, resp



$$\sim \mathcal{N}\left(\mu, \boldsymbol{\Sigma}\right)$$

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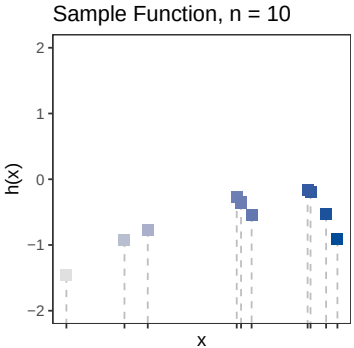
$$\mathbf{f} = f(\boldsymbol{X}) = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- $\mathbf{m}$ and $\mathbf{K}$ are calculated by a mean and cov function, resp



$$\sim \mathcal{N}\left(\mu, \boldsymbol{\Sigma}\right)$$

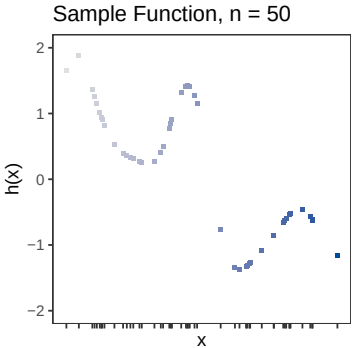
MARGINALIZATION PROPERTY

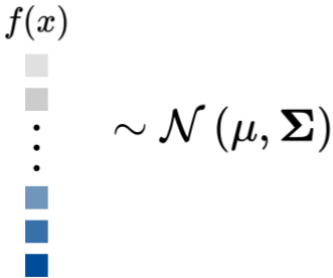
- Consider finite set of inputs $\boldsymbol{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\} \subset \mathcal{X}$

- Distinguishing feature of GPs: for any such $\boldsymbol{X}$,

$$\mathbf{f} = f(\boldsymbol{X}) = \left[f\left(\mathbf{x}^{(1)}\right), \ldots, f\left(\mathbf{x}^{(n)}\right)\right] \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$$

- $\mathbf{m}$ and $\mathbf{K}$ are calculated by a mean and cov function, resp





DEFINITION: GAUSSIAN PROCESS

- Formally,

$$f \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$$

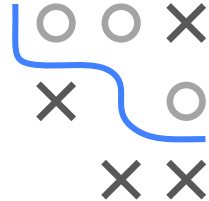
  iff for any finite set of inputs $\boldsymbol{X} \subset \mathcal{X}$,

$$f(\boldsymbol{X}) \sim \mathcal{N}(m(\boldsymbol{X}), k(\boldsymbol{X}, \boldsymbol{X}))$$

- Depends on **mean function** $m : \mathcal{X} \to \mathbb{R}$ and **covariance function** $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_0^+$

- For specific set $\boldsymbol{X}$ we get (deterministic) vectors / mats
  - $\mathbf{m} = m(\boldsymbol{X}) = [m(\mathbf{x}^{(1)}), \dots, m(\mathbf{x}^{(n)})]$
  - $\mathbf{K} = k(\boldsymbol{X}, \boldsymbol{X}) = (k(\mathbf{x}, \mathbf{x}'))_{\mathbf{x},,\mathbf{x}' \in X}$
  - $\mathbf{f} \sim \mathcal{N}(\mathbf{m}, \mathbf{K})$

EXISTENCE THEOREM



- For **any**
  - state space $\mathcal{X}$,
  - mean function $m : \mathcal{X} \to \mathbb{R}$,
  - covariance function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}_0^+$,

  there **exists** $\mathcal{GP}(m(\cdot), k(\cdot,\cdot))$ s.t. $\forall \mathbf{x}, \mathbf{x}' \in \mathcal{X}$

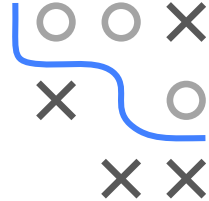
$$\begin{aligned} \mathbb{E}(f(\mathbf{x})) &= m(\mathbf{x}) \\ \mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) &= k\left(\mathbf{x}, \mathbf{x}'\right) \end{aligned}$$

  and $f(\boldsymbol{X}) \sim \mathcal{N}(m(\boldsymbol{X}), k(\boldsymbol{X}, \boldsymbol{X}))$ for any $\boldsymbol{X} \subset \mathcal{X}$

- Intuition: for any mean & cov structure we prescribe, there is a corresponding GP to generate **f** with these characteristics

- Version of Kolmogorov consistency theorem
  $\Rightarrow$ proof: ▸ Grimmett and Stirzaker 2001 (Thm. 8.6.3)

IMPLICATIONS OF EXISTENCE THM



- GPs completely specified by their mean & cov function

$$
\begin{aligned}
m(\mathbf{x}) &= \mathbb{E}[f(\mathbf{x})] \\
k(\mathbf{x}, \mathbf{x}') &= \mathrm{Cov}(f(\mathbf{x}), f(\mathbf{x}')) = \mathbb{E}\left[(f(\mathbf{x}) - \mathbb{E}[f(\mathbf{x})])\,(f(\mathbf{x}') - \mathbb{E}[f(\mathbf{x}')])\right]
\end{aligned}
$$

- For now, we consider zero-mean GPs with $m(\mathbf{x}) \equiv \mathbf{0}$
  $\Rightarrow$ common, not necessarily drastic assumption

- Denote by $\mathcal{GP}(\mathbf{0}, k(\cdot,\cdot)) \Rightarrow$ properties mainly governed by $k(\cdot,\cdot)$

- By virtue of existence thm: sampling from GP priors gives us random functions with our properties of choice

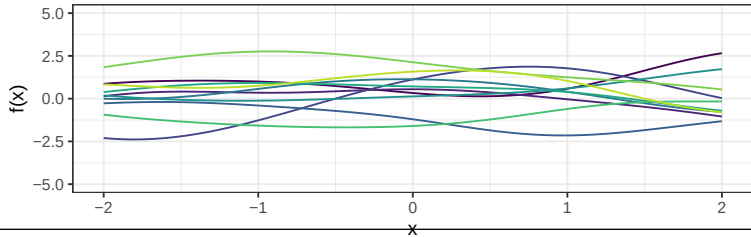
SAMPLING FROM GAUSSIAN PROCESS PRIORS

- Example: $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot,\cdot))$ with cov function

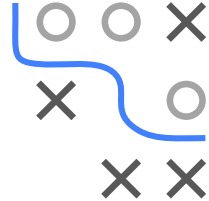
$$k\left(\mathbf{x}, \mathbf{x}'\right) = \exp\left(-\tfrac{1}{2}\|\mathbf{x} - \mathbf{x}'\|^2\right)$$

- To visualize sample functions,
  - choose high number n of (equidistant) points $\boldsymbol{X} = \left\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n)}\right\}$
  - compute $\mathbf{K} = k(\boldsymbol{X}, \boldsymbol{X})$ from all pairs $\mathbf{x}^{(i)}, \mathbf{x}^{(j)} \in \boldsymbol{X}$
  - draw $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$

- 10 randomly drawn functions (note 0 mean)



SAMPLING FROM GAUSSIAN PROCESS PRIORS



- Example: 2D Brownian motion
- $f \sim \mathcal{GP}(\mathbf{0}, k(\cdot, \cdot))$ with cov function

$$k\left(\mathbf{x}, \mathbf{x}'\right) = \min(\mathbf{x}, \mathbf{x}')$$

- TODO LW: plot